

WS22: Mock 2: Specimen Paper 4, Q1 and Q2

Time: 1 h 15 min

Name: **SOLUTION** ()

Class: **26S**

Date:

1 This experiment considers the forces on a wooden cylinder.

(a) You are provided with a wooden cylinder with a spring attached.

The distance L between the centre of the hole with the string and the end of the cylinder is shown in Fig. 1.1.

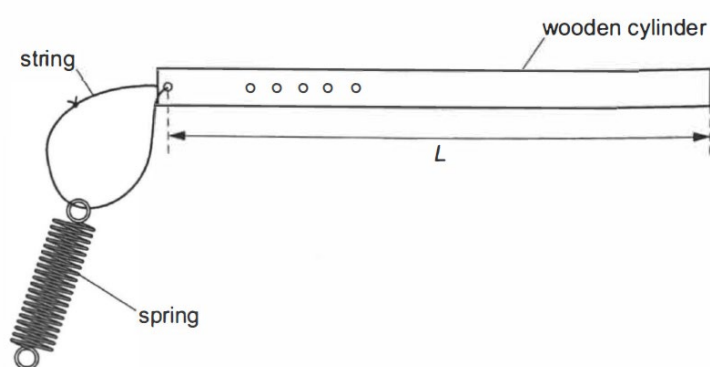


Fig. 1.1

The length of the unstretched spring is S , as shown in Fig. 1.2.

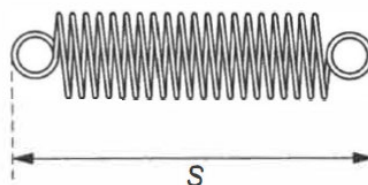


Fig. 1.2

Measure and record L and S .

$L = 59.0 \text{ cm}$

$S = 5.2 \text{ cm}$ [1]

Note: reject $L = 60.0 \text{ cm}$

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- (b) Set up the apparatus as shown in Fig. 1.3.

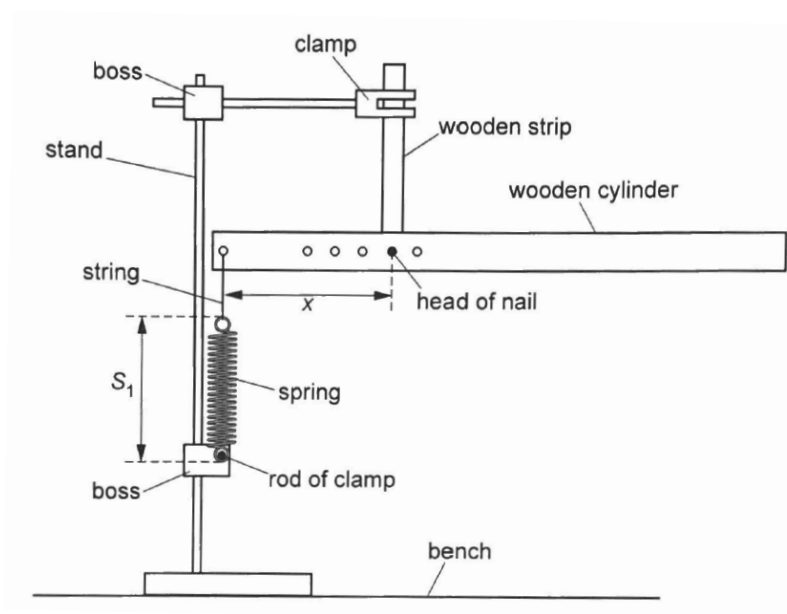


Fig. 1.3

Place the nail through one of the holes in the wooden cylinder.

Adjust the apparatus until the spring and the wooden strip are vertical and the wooden cylinder is horizontal.

The distance x between the hole with the string and the hole with the nail is shown in Fig. 1.3.

The length of the stretched spring is S_1 .

- (i) Measure and record x and S_1 .

$$x = \underline{21.0 \text{ cm}}$$

$$S_1 = \underline{6.2 \text{ cm}} \quad [1]$$

- (ii) Calculate e and n , where

$$e = S_1 - S \quad \text{and} \quad n = \frac{x}{L}.$$

$$\begin{aligned} e &= 6.2 - 5.2 \\ &= 1.0 \text{ cm} \end{aligned}$$

$$e = \underline{1.0 \text{ cm}}$$

$$\begin{aligned} n &= \frac{21.0}{59.0} \\ &= 0.356 \end{aligned}$$

$$n = \underline{0.356} \quad [1]$$

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(c) Vary distance x .

Adjust the apparatus until the spring and the wooden strip are vertical and the wooden cylinder is horizontal.

Repeat **1(b)(i)** and **1(b)(ii)**.

Present your results clearly.

x / cm	s_1 / cm	n	e / cm	$1/n$
21.0	6.2	0.356	1.0	2.81
18.1	7.6	0.307	2.4	3.26
15.1	9.5	0.256	4.3	3.91
12.1	12.5	0.205	7.3	4.88
9.1	17.4	0.15	12.2	6.5

[3]

(d) The quantities e and n are related by the equation

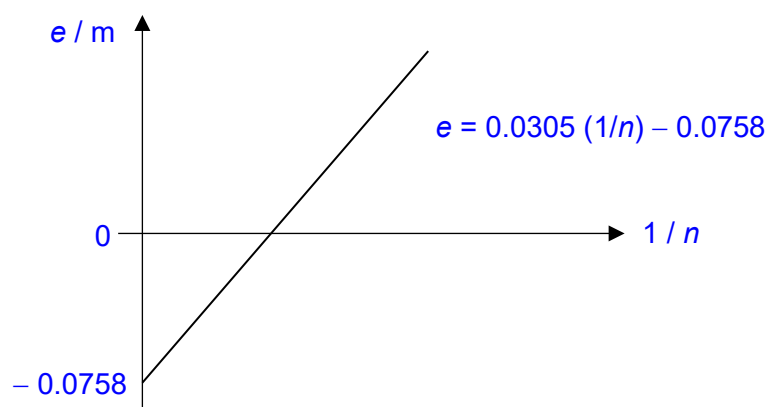
$$e = \frac{P}{n} + Q$$

where P and Q are constants.

(i) Plot a suitable graph using a spreadsheet to determine P and Q .

Sketch the graph in the space below. Write down the equation of the trendline.

Plot the graph of e / m vs $1 / n$ where the gradient is P and the vertical intercept is Q .



$P =$ 0.0305 m

$Q =$ -0.0758 m

[3]

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- (ii) Theory suggests that

$$P = \frac{Mg}{2k}$$

where g is 9.81 N kg^{-1} , M is the mass of the wooden cylinder and k is the spring constant of the spring.

Mass M is given on the card.

Calculate k .

$$k = \frac{Mg}{2P} = \frac{0.13522 \times 9.81}{2 \times 0.03050} = 21.7 \text{ N m}^{-1}$$

$$k = \underline{21.7} \text{ N m}^{-1} \quad [1]$$

- (e) Suggest a theoretical value for e when n is equal to 0.5. Explain your answer.

When $n = 0.5 \Rightarrow x = \frac{1}{2}L$, this means that the ruler is supported at its centre of gravity.

So, the extension of the spring, $e = 0$.

[2]

[Total: 12]

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- 2** This investigation considers the current in an RC circuit when a capacitor of capacitance C charges and discharges.

Set up the circuit shown in Figure 2.1, taking care to connect the capacitor the right way round. Switches S_1 and S_2 are initially unconnected.

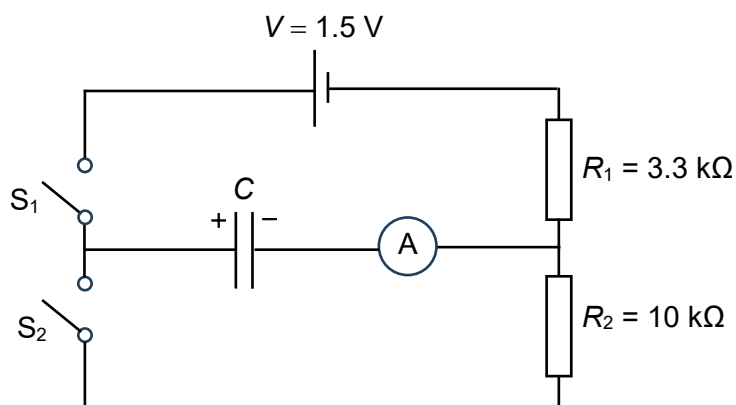


Figure. 2.1

- (a) (i)** Close switch S_1 and wait about 20 s.

Record the reading I on the ammeter.

$$I = \underline{0.4 \mu A}$$

[1]

- (ii)** Open switch S_1 and close switch S_2 and immediately start the stopwatch.

Record I every 10 s from time $t = 0$ to $t = 50$ s. Tabulate your results.

You may need several attempts before you have a complete set of results.

t / s	$I_1 / \mu\text{A}$	$I_2 / \mu\text{A}$	$I_{\text{ave}} / \mu\text{A}$
0.0	153.5	152.8	153.2
10.0	63.7	63.4	63.6
20.0	19.2	18.8	19.0
30.0	6.3	6.0	6.2
40.0	2.2	2.0	2.1
50.0	0.8	0.6	0.7

[2]

- (iii) Describe **one** problem in taking the readings.

.....
There is a short time interval between looking at the stopwatch and then the ammeter. Since
.....
the current is decreasing quickly, the recorded current I is smaller than the true value. This
..... [1]
introduces a systematic error in the current readings.
.....

- (b) An engineer wants to test the capacitance of an unmarked capacitor in order to design a circuit to filter noise from received signals.

To determine the unknown value of C , the engineer uses an RC circuit similar to that in Figure 2.1 but with different values of V , R_1 and R_2 .

The engineer discharges the fully charged capacitor through resistor R_2 .

The current in the circuit decreases exponentially with discharging time t from an initial value I_0 in the form:

$$I = I_0 e^{-\frac{t}{\tau}}$$

where τ is the time constant of the discharging circuit.

The engineer's discharging data is recorded in the digital file: datasetQ2b.xls.

Imagine you are the engineer.

- (i) Use the engineer's data to estimate the charge Q_1 that flows through the circuit during discharging.

Q_1 in the range 0.00250–0.00270 mC

$Q_1 =$ 0.00252 mC [1]

- (ii) Explain how you obtained the value of Q_1 .

.....
The charge Q_1 that flows through the circuit during discharging is the area under the
.....
current-time graph from $t = 0$ to $t = \infty$ (as $Q_1 = \int_0^{\infty} I dt$).
.....
This is obtained by applying the trapezium rule to the current-time curve in Excel. [1]

Note: $Q \neq It$ because current is not constant.

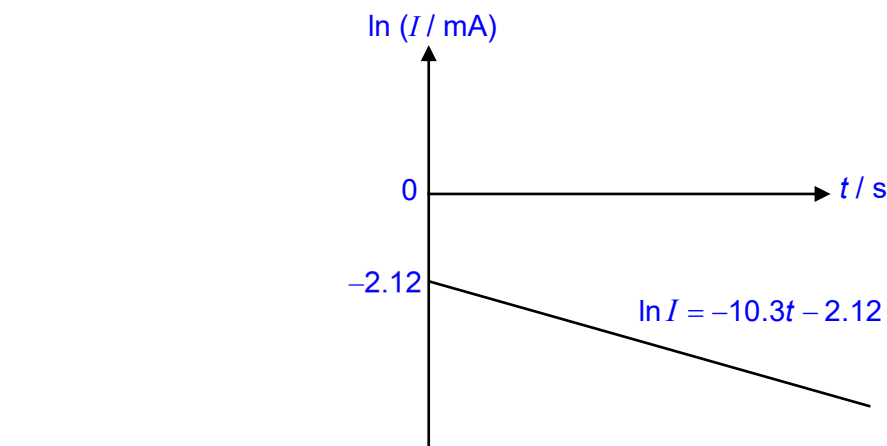
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- (iii) Plot a suitable graph using a spreadsheet to determine the time constant τ of the engineer's discharging circuit.

Sketch the graph in the space below. Write down the equation of the trendline.

Given that $I = I_0 e^{-\frac{t}{\tau}} \Rightarrow \ln I = \ln \left(I_0 e^{-\frac{t}{\tau}} \right) = \ln I_0 - \frac{t}{\tau}$

Plot a graph of $\ln I$ against t , where the gradient is $-\frac{1}{\tau}$ and the vertical intercept is $\ln I_0$



gradient of line $= -\frac{1}{\tau} = -10.3$

$\therefore \tau = 0.0971 \text{ s}$

τ in the range 0.0950–0.0980 s

$\tau = \underline{0.0971 \text{ s}}$ [2]

- (c) The electric potential energy U stored in a capacitor of known capacitance C is given by

$$U = \frac{1}{2} CV^2$$

where V is the potential difference (p.d.) across the capacitor.

Plan an investigation to determine the difference between the energy stored on the fully charged capacitor in Figure 2.1 and the work done by the cell in fully charging the capacitor.

You are not required to do this experiment.

Include in your plan:

- a circuit diagram
- the procedure to be followed
- how the work done by the cell is determined while the capacitor is charging
- any precautions that should be taken to improve the accuracy and/or the safety of the investigation.

You may suggest the use of any additional apparatus commonly found in a school physics laboratory.

Diagram

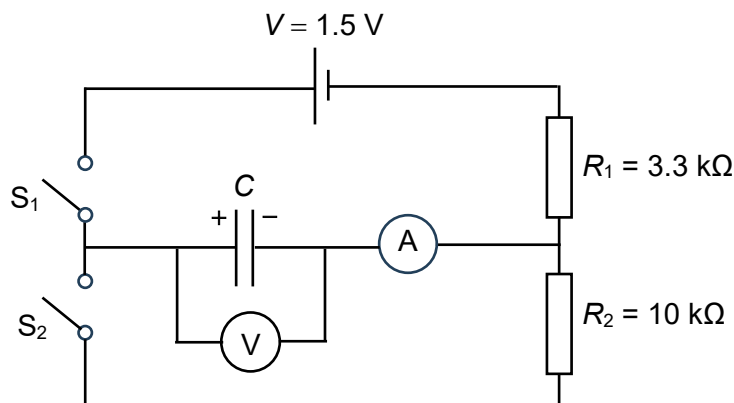


Figure. 2.2

Procedure

1. Set up the circuit as shown in Figure 2.2. Open switch S_1 open and close switch S_2 until the capacitor is fully discharged (when both ammeter and voltmeter readings are zero). Open switch S_2 .
2. Close switch S_1 to charge the capacitor and simultaneously start a stopwatch. Record the ammeter reading I and stopwatch reading t at fixed time intervals until the current falls to zero.
3. Repeat steps 1 to 3 to obtain a second set of currents (I) at the same times (t) to determine the average current $\langle I \rangle$.
4. Record the voltmeter reading V across the capacitor when it is fully charged (or the p.d. / e.m.f. across the cell).

** If a data-logger is used to record the current, relevant details must be provided.

Analysis

1. Plot the graph of $\langle I \rangle$ against t . The charge stored Q is the area under the graph, which can be found by applying the trapezium rule using a spreadsheet.
2. The work done W by the cell in fully charging the capacitor to a potential difference V is $W = QV$. Hence, the difference ΔE between the work done W by the cell and the energy U stored in the capacitor is

$$\Delta E = W - U = QV - \frac{1}{2}CV^2 = QV - \frac{1}{2}QV = \frac{1}{2}QV$$

Reliability

1. Before repeating the experiment, fully discharge the capacitor by checking that the current reaches zero when switch S_2 is closed and switch S_1 is opened.
2. Select a suitable value of R_1 such that the charging is not too rapid, so that the experimenter is able to record both readings of t and I accurately.

Safety Precautions

1. To prevent damage to the capacitor, connect the positive terminal of the capacitor to the positive terminal of the battery.
2. To avoid an electrical shock or a large current during discharging, resistor R_2 must always be sufficiently large. Or, the capacitor must be discharged before storage to prevent accidental electrical shock.

[6]
[Total: 14]

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s/n	Answer
1	Circuit diagram showing voltmeter across capacitor.
2	Measure current using ammeter I and time t using stopwatch.
3	Workable method to determine when capacitor is fully charged and the final p.d. V across it.
4	Workable method to determine final charge Q stored in capacitor and work done W by cell.
5	Valid conclusion on difference between W and U .
6	Reference to a valid safety precaution, e.g., care to discharge capacitor to avoid electric shock, care to avoid short-circuit/low resistance and overheating of the wires. Averaging normally not awarded marks. OR Reference to a valid reliability precaution, e.g., slow discharge to record time and current accurately, fully discharging capacitor before data collection.